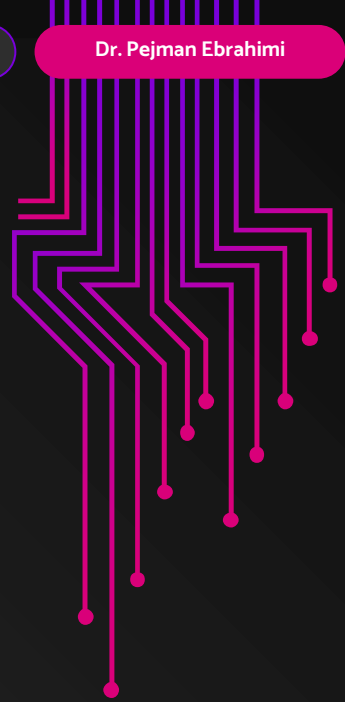


System Analysis and Design

Lecture 1 · Setting up your development environment



Meet your instructor

Dr. Pejman Ebrahimi is a Postdoctoral Researcher in Data Science and AI at the University of Liechtenstein, with a PhD from MATE University, Budapest. He has spent ten years building machine-learning models, data pipelines and autonomous AI agents for industry and academia. Selected among global young researchers for the 8th Lindau Nobel Laureate Meeting (August 26-30, 2025).



Today's roadmap

01

Python & Git Bash

The language and the terminal
you will use

02

PyCharm

Download, install, and register it
on PATH

03

First project

Create a project and run your
first file

04

Verify & fix

Confirm every tool answers
before week two

Three tools, one workflow



Python 3.14

The language itself – the interpreter that runs your code.



Git Bash

A Unix-style terminal for Windows, with Git included.



PyCharm

The IDE where you write, run and debug all semester.

Install Python in six steps

Download

python.org → Downloads →
latest 3.12

Tick the box

Add python.exe to PATH –
before anything else

Install now

Accept the defaults and the
admin prompt

Reopen terminal

A shell only reads PATH
when it starts

Verify

python --version answers
Python 3.14.x

Check pip

pip --version confirms the
package manager

Git Bash: your terminal

Why Git Bash?

Windows ships PowerShell, but nearly every tutorial you read uses Unix commands. Git Bash gives you ls, cd, grep and git. On macOS or Linux you only need Git itself.

• How to install it

git-scm.com → Download for Windows → run the installer → keep the defaults → confirm with git --version.

Install PyCharm step by step

Download

Community Edition, free,
from jetbrains.com

Run installer

Accept the default
install folder

Options

Tick: shortcut, .py, Add
bin to PATH

Restart

A reboot activates the
new PATH entry

First launch

Accept terms, pick a
theme, skip plugins

Licence

Professional is free with
a uni.li address

Do not forget to add it to PATH

PyCharm

Tick "Add bin folder to
PATH"

Python

Tick "Add python.exe to
PATH"

Why bother

Otherwise: command
not found

Forgot?

Re-run installer →
Modify → tick

Create your first project

Welcome screen → New
Project

Base interpreter = Python
3.14

New Project

Interpreter

Name it

Call it sad_lecture1, no
spaces

Create

PyCharm opens the editor

Create a file and run it



Right-click

Right-click the project folder
→ New → Python File



Name it

Type hello and press Enter –
PyCharm adds .py



Run it

Write your code, then press
Shift + F10

Check your setup before next week

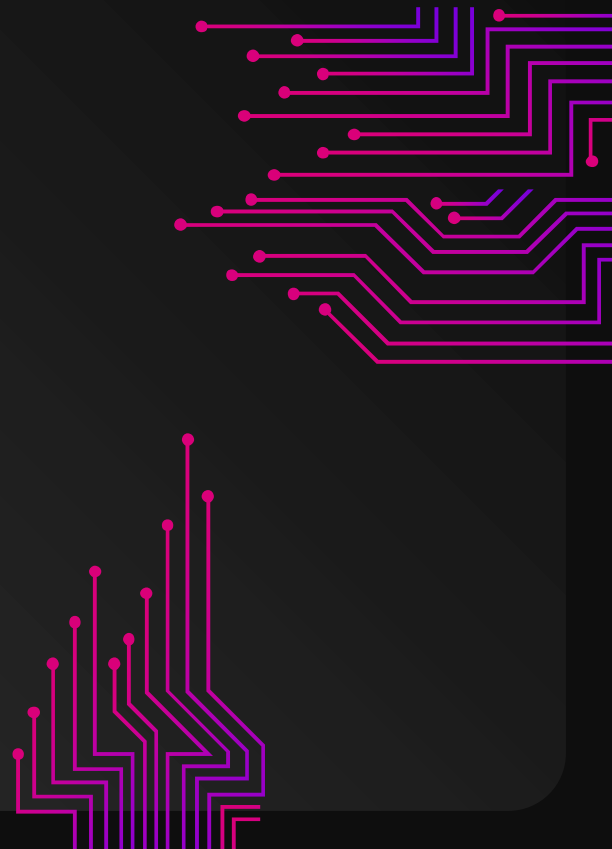
Check	Type this in the terminal	You should see
Python	<code>python --version</code>	Python 3.14.x
Packages	<code>pip --version</code>	pip 24.x from ...
Git	<code>git --version</code>	git version 2.4x.x
PATH (Win)	<code>where python</code>	C:\...\python.exe
PATH (mac)	<code>which python3</code>	/usr/local/bin/python3
Git Bash	<code>ls</code>	your files
First run	<code>python hello.py</code>	Hello from Pejman!

When something goes wrong

Four problems cause almost every week-one question:

- **'python' is not recognized** – PATH was never ticked. Re-run the installer, choose Modify.
- **The wrong version answers** – an older Python sits earlier in PATH. Remove it.
- **No interpreter in PyCharm** – Settings → Project → Python Interpreter → Add.

Still stuck? Send me the exact command and the exact error text.



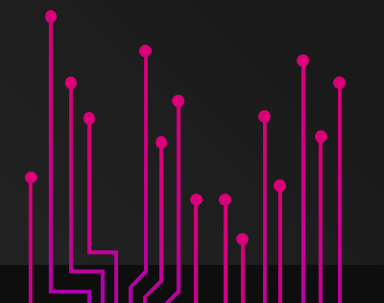
Before the next lecture

Do this week

Install all three tools on the laptop you bring to class. Create `sad-lecture1`, add `hello.py`, run it, screenshot the output.

Bring to class

A working laptop, your screenshot, and one question about whatever refused to work.



Thanks!

Questions are always welcome

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